

## Additional file 1

### Supplementary methods, tables, and figures

Supplement to:

#### **Multi-Modal Molecular Representation Learning Prioritizes Class I HDAC Inhibitors for Pan-Cancer Transcriptomic Reversal**

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**Evidence policy.** This supplement distinguishes predicted reversal, measured LINCS transcriptomic reversal, biological context, and structural sensitivity. Measured reversal is not a viability assay; reversal-associated genes are not asserted direct drug targets; DepMap knockout effects are not compound responses; and docking is not binding validation. The 24 tables below are compact human-readable views. Full, filterable rows and source fields are supplied in Additional file 2.

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## 1 Scope, corrections, and candidate policy

The revision uses three completed computational evidence packages: (E1) official-condition measured LINCS comparison and prediction–measurement concordance; (E2) biological reanalysis after candidate roles were frozen, including networks, pathways, and DepMap Public 26Q1; and (E3) controlled zinc-aware docking with crystallographic redocking, a designed zinc-chelation decoy, and receptor sensitivity. A direct external drug-sensitivity package was conditional on candidate-matched public data and was not performed because such data were unavailable for TC-H-106. No new wet-lab experiment is claimed.

The candidate hierarchy was frozen before network, pathway, dependency, and docking interpretation. Mocetinostat is the core computational candidate; NCH-51 is secondary; TC-H-106 is exploratory/original-method-sensitive; RG2833 is prediction-only because no measured LINCS signature was found; and Tianeptinaline/BG-1010 is excluded from primary inference because official identifier resolution exposed a name–structure conflict. Belinostat, PCI-24781, and Panobinostat provide class support; Entinostat and Vorinostat are reference controls.

The revised package removes simulated TCGA survival and stage analyses, unsupported tumor-mutation-burden conclusions, proxy DeepCE/ChemCPA claims, unsupported legacy ablation values, GDSC surrogate “validation,” direct-target language for reversed genes, and MMFF94 energy claims as thermodynamic evidence. A pre-correction HDAC holdout is retained only as provenance; all model inference uses the corrected annotation-defined 1,856-profile, 30-structure holdout.

## 2 Dataset, model scope, and leakage-aware evaluation

Table S1 separates signatures, perturbagen tokens, nominal conditions, canonical structures, and cell lines. This distinction prevents the 55,695 modeling rows from being described as unique compounds or biological replicates.

Table S1: Composition of the quality-controlled LINCS modeling dataset and frozen screening library. Level 5 signatures are not unique compounds or raw biological replicates.

| Item                                            | Count  | Interpretation                                                  |
|-------------------------------------------------|--------|-----------------------------------------------------------------|
| Level 5 signatures                              | 55,695 | Modeling rows after quality control                             |
| Signature-ID perturbagen tokens                 | 13,314 | Includes instance suffixes                                      |
| Base perturbagen identifiers                    | 11,527 | Normalized BRD identifiers                                      |
| Unique canonical structures                     | 11,445 | Valid RDKit structures                                          |
| Cell lines                                      | 65     | Across the modeling dataset                                     |
| Perturbagen-token-cell combinations             | 39,266 | Descriptive combinations                                        |
| Perturbagen-token-cell-time-dose conditions     | 52,834 | Nominal conditions                                              |
| Additional signatures beyond nominal conditions | 2,861  | Plate/batch-level Level 5 entries                               |
| Expression genes                                | 12,328 | Shared disease/perturbation feature space                       |
| Screened canonical compounds                    | 28,477 | Frozen screening library; distinct from the modeling structures |

Both neural predictors are chemical-structure-only models. They do not receive cell-line identity, dose, or time. Their output is therefore a structure-conditioned central tendency across training conditions, not a condition-specific response. Leave-cell-line evaluation measures agreement with profiles from held-out cellular backgrounds under this formulation; it does not show that the models learned cell-specific response rules.

Table S2: Frozen model, optimization, and interpretation settings.

| Item                       | Frozen specification                                                                            |
|----------------------------|-------------------------------------------------------------------------------------------------|
| Prediction input           | Canonical chemical structure only; no cell-line, dose, or time covariates                       |
| Target                     | 12,328-gene LINCS Level 5 perturbation vector                                                   |
| Atom stream                | 43-dimensional element identity; 3 Transformer layers; 512 pooled dimensions; no bond adjacency |
| Fingerprint stream         | 1,024-bit radius-2 Morgan fingerprint; 512-dimensional projection                               |
| Dual-stream parameters     | 38,682,152                                                                                      |
| Fingerprint MLP parameters | 28,415,016                                                                                      |
| Optimization               | MSE; AdamW; learning rate 2e-4; weight decay 1e-4; batch 128                                    |
| Stopping                   | Maximum 30 epochs; patience 5; checkpoint by validation MSE                                     |
| Benchmark seeds            | Pair split 1–5; other splits 1–3; grouping seed 42                                              |
| Primary screening metrics  | signed wTRS and negative Spearman disease–perturbation correlation                              |
| Stability configurations   | 48 primary; 72 only when legacy wTRS is added as sensitivity                                    |

The split audit first checks signature identifiers and then distinguishes exact-structure and scaffold exposure. Zero scaffold overlap is required only for the scaffold split. The corrected annotation-defined HDAC holdout requires zero exact HDAC structure overlap; 11 scaffolds remain shared and are reported rather than mislabeled as leakage.

Table S3: Leakage and structural-overlap audit. Exact-structure overlap is zero for leave-drug, scaffold, and corrected annotation-defined HDAC evaluations; the scaffold column is the directly recorded Murcko overlap.

| Split                             | Train n | Test n | Train struct. | Test struct. | Sig. ovlp. | Scaffold ovlp. |
|-----------------------------------|---------|--------|---------------|--------------|------------|----------------|
| Drug–cell pair                    | 47,376  | 8,319  | 10,582        | 3,827        | 0          | 2,489          |
| Leave drug out                    | 47,728  | 7,967  | 9,728         | 1,717        | 0          | 686            |
| Leave cell line out               | 50,520  | 5,175  | 11,170        | 2,826        | 0          | 1,914          |
| Scaffold split                    | 47,851  | 7,844  | 9,831         | 1,614        | 0          | 0              |
| Corrected annotation-defined HDAC | 53,839  | 1,856  | 11,415        | 30           | 0          | 11             |

Table S4: Generalization summary as mean (seed SD). Corrected 1,856-profile HDAC runs replace stale pre-correction summaries.

| Split                                     | Model           | Seeds | Test | MSE           | MAE           | $R^2$         | Pearson       | Spearman      |
|-------------------------------------------|-----------------|-------|------|---------------|---------------|---------------|---------------|---------------|
| Corrected annotation-defined HDAC holdout | Dual stream     | 3     | 1856 | 1.326 (0.026) | 0.839 (0.005) | 0.131 (0.017) | 0.378 (0.009) | 0.345 (0.010) |
| Corrected annotation-defined HDAC holdout | Fingerprint MLP | 3     | 1856 | 1.333 (0.015) | 0.841 (0.004) | 0.127 (0.010) | 0.379 (0.007) | 0.344 (0.006) |
| Leave cell line out                       | Dual stream     | 3     | 5175 | 0.942 (0.004) | 0.714 (0.000) | 0.121 (0.003) | 0.266 (0.010) | 0.243 (0.010) |
| Leave cell line out                       | Fingerprint MLP | 3     | 5175 | 0.941 (0.007) | 0.713 (0.002) | 0.122 (0.006) | 0.272 (0.010) | 0.249 (0.010) |
| Leave drug out                            | Dual stream     | 3     | 7967 | 0.919 (0.003) | 0.717 (0.001) | 0.052 (0.003) | 0.188 (0.000) | 0.166 (0.001) |
| Leave drug out                            | Fingerprint MLP | 3     | 7967 | 0.919 (0.001) | 0.716 (0.001) | 0.052 (0.001) | 0.191 (0.003) | 0.170 (0.003) |
| Drug-cell pair                            | Dual stream     | 5     | 8319 | 0.882 (0.004) | 0.704 (0.003) | 0.108 (0.004) | 0.259 (0.009) | 0.237 (0.009) |
| Drug-cell pair                            | Fingerprint MLP | 5     | 8319 | 0.881 (0.001) | 0.703 (0.001) | 0.110 (0.002) | 0.263 (0.005) | 0.241 (0.005) |
| Scaffold split                            | Dual stream     | 3     | 7844 | 0.973 (0.000) | 0.738 (0.002) | 0.062 (0.000) | 0.185 (0.006) | 0.163 (0.007) |
| Scaffold split                            | Fingerprint MLP | 3     | 7844 | 0.971 (0.002) | 0.736 (0.001) | 0.064 (0.002) | 0.186 (0.001) | 0.165 (0.002) |

Table S5: All 34 frozen model runs. Runtime includes training and evaluation on the recorded RTX 4090 environment.

| Split          | Model       | Seed | Test  | Epoch | Runtime s | MSE   | MAE   | $R^2$ | Pearson | Spearman |
|----------------|-------------|------|-------|-------|-----------|-------|-------|-------|---------|----------|
| Corrected HDAC | Dual        | 1    | 1,856 | 4     | 260.5     | 1.320 | 0.837 | 0.135 | 0.381   | 0.348    |
| Corrected HDAC | Dual        | 2    | 1,856 | 5     | 282.7     | 1.355 | 0.845 | 0.112 | 0.385   | 0.353    |
| Corrected HDAC | Dual        | 3    | 1,856 | 8     | 365.0     | 1.305 | 0.834 | 0.145 | 0.368   | 0.334    |
| Corrected HDAC | Fingerprint | 1    | 1,856 | 2     | 54.1      | 1.347 | 0.844 | 0.118 | 0.387   | 0.351    |
| Corrected HDAC | Fingerprint | 2    | 1,856 | 5     | 72.5      | 1.336 | 0.842 | 0.125 | 0.373   | 0.338    |
| Corrected HDAC | Fingerprint | 3    | 1,856 | 4     | 70.3      | 1.316 | 0.837 | 0.138 | 0.378   | 0.344    |
| Leave cell     | Dual        | 1    | 5,175 | 6     | 209.5     | 0.938 | 0.713 | 0.125 | 0.259   | 0.236    |
| Leave cell     | Dual        | 2    | 5,175 | 7     | 212.4     | 0.945 | 0.713 | 0.118 | 0.261   | 0.238    |
| Leave cell     | Dual        | 3    | 5,175 | 15    | 394.8     | 0.942 | 0.714 | 0.121 | 0.277   | 0.254    |
| Leave cell     | Fingerprint | 1    | 5,175 | 13    | 78.9      | 0.937 | 0.712 | 0.125 | 0.277   | 0.254    |
| Leave cell     | Fingerprint | 2    | 5,175 | 7     | 58.4      | 0.949 | 0.715 | 0.115 | 0.261   | 0.238    |
| Leave cell     | Fingerprint | 3    | 5,175 | 15    | 91.5      | 0.937 | 0.711 | 0.126 | 0.279   | 0.256    |
| Leave drug     | Dual        | 1    | 7,967 | 2     | 132.4     | 0.916 | 0.716 | 0.055 | 0.187   | 0.165    |
| Leave drug     | Dual        | 2    | 7,967 | 4     | 163.5     | 0.921 | 0.718 | 0.049 | 0.188   | 0.166    |
| Leave drug     | Dual        | 3    | 7,967 | 5     | 182.0     | 0.921 | 0.717 | 0.050 | 0.188   | 0.167    |
| Leave drug     | Fingerprint | 1    | 7,967 | 5     | 48.5      | 0.918 | 0.716 | 0.053 | 0.195   | 0.173    |
| Leave drug     | Fingerprint | 2    | 7,967 | 2     | 36.7      | 0.919 | 0.715 | 0.052 | 0.188   | 0.167    |
| Leave drug     | Fingerprint | 3    | 7,967 | 5     | 51.7      | 0.920 | 0.716 | 0.051 | 0.191   | 0.169    |
| Pair           | Dual        | 1    | 8,319 | 13    | 327.2     | 0.880 | 0.702 | 0.111 | 0.268   | 0.246    |
| Pair           | Dual        | 2    | 8,319 | 5     | 191.6     | 0.888 | 0.707 | 0.102 | 0.248   | 0.225    |
| Pair           | Dual        | 3    | 8,319 | 14    | 336.4     | 0.877 | 0.701 | 0.113 | 0.268   | 0.245    |
| Pair           | Dual        | 4    | 8,319 | 8     | 237.8     | 0.881 | 0.704 | 0.109 | 0.256   | 0.233    |
| Pair           | Dual        | 5    | 8,319 | 10    | 278.0     | 0.885 | 0.705 | 0.106 | 0.258   | 0.236    |
| Pair           | Fingerprint | 1    | 8,319 | 14    | 86.7      | 0.881 | 0.703 | 0.109 | 0.268   | 0.246    |
| Pair           | Fingerprint | 2    | 8,319 | 11    | 71.9      | 0.879 | 0.702 | 0.112 | 0.263   | 0.241    |
| Pair           | Fingerprint | 3    | 8,319 | 6     | 55.0      | 0.883 | 0.704 | 0.108 | 0.254   | 0.232    |
| Pair           | Fingerprint | 4    | 8,319 | 14    | 87.5      | 0.881 | 0.703 | 0.110 | 0.266   | 0.244    |
| Pair           | Fingerprint | 5    | 8,319 | 10    | 72.7      | 0.880 | 0.702 | 0.111 | 0.263   | 0.240    |
| Scaffold       | Dual        | 1    | 7,844 | 2     | 135.6     | 0.973 | 0.735 | 0.062 | 0.179   | 0.156    |
| Scaffold       | Dual        | 2    | 7,844 | 4     | 167.6     | 0.974 | 0.739 | 0.061 | 0.191   | 0.170    |
| Scaffold       | Dual        | 3    | 7,844 | 3     | 152.6     | 0.974 | 0.739 | 0.061 | 0.186   | 0.163    |
| Scaffold       | Fingerprint | 1    | 7,844 | 6     | 56.7      | 0.974 | 0.736 | 0.062 | 0.187   | 0.165    |
| Scaffold       | Fingerprint | 2    | 7,844 | 3     | 45.2      | 0.969 | 0.735 | 0.066 | 0.185   | 0.163    |
| Scaffold       | Fingerprint | 3    | 7,844 | 3     | 44.7      | 0.971 | 0.736 | 0.064 | 0.186   | 0.166    |

Table S6: Paired model differences. Positive values favor the dual stream; all intervals include zero except the candidate-specific PCI-24781 comparison reported separately.

| Split          | Metric   | Seeds | Dual advantage | SD    | 95% low | 95% high |
|----------------|----------|-------|----------------|-------|---------|----------|
| Pair           | MSE      | 5     | -0.001         | 0.006 | -0.009  | 0.006    |
| Pair           | MAE      | 5     | -0.001         | 0.003 | -0.005  | 0.003    |
| Pair           | R2       | 5     | -0.001         | 0.006 | -0.009  | 0.006    |
| Pair           | Pearson  | 5     | -0.003         | 0.011 | -0.017  | 0.010    |
| Pair           | Spearman | 5     | -0.003         | 0.011 | -0.018  | 0.011    |
| Leave drug     | MSE      | 3     | -0.001         | 0.002 | -0.006  | 0.005    |
| Leave drug     | MAE      | 3     | -0.002         | 0.001 | -0.005  | 0.002    |
| Leave drug     | R2       | 3     | -0.001         | 0.002 | -0.006  | 0.005    |
| Leave drug     | Pearson  | 3     | -0.003         | 0.004 | -0.013  | 0.006    |
| Leave drug     | Spearman | 3     | -0.004         | 0.004 | -0.013  | 0.005    |
| Leave cell     | MSE      | 3     | -0.001         | 0.004 | -0.012  | 0.010    |
| Leave cell     | MAE      | 3     | -0.001         | 0.002 | -0.006  | 0.005    |
| Leave cell     | R2       | 3     | -0.001         | 0.004 | -0.011  | 0.009    |
| Leave cell     | Pearson  | 3     | -0.007         | 0.010 | -0.031  | 0.018    |
| Leave cell     | Spearman | 3     | -0.006         | 0.010 | -0.031  | 0.018    |
| Scaffold       | MSE      | 3     | -0.002         | 0.003 | -0.009  | 0.004    |
| Scaffold       | MAE      | 3     | -0.002         | 0.003 | -0.009  | 0.004    |
| Scaffold       | R2       | 3     | -0.002         | 0.003 | -0.009  | 0.004    |
| Scaffold       | Pearson  | 3     | -0.001         | 0.007 | -0.018  | 0.016    |
| Scaffold       | Spearman | 3     | -0.002         | 0.008 | -0.022  | 0.018    |
| Corrected HDAC | MSE      | 3     | 0.006          | 0.023 | -0.051  | 0.064    |
| Corrected HDAC | MAE      | 3     | 0.002          | 0.005 | -0.010  | 0.014    |
| Corrected HDAC | R2       | 3     | 0.004          | 0.015 | -0.034  | 0.042    |
| Corrected HDAC | Pearson  | 3     | -0.001         | 0.011 | -0.030  | 0.027    |
| Corrected HDAC | Spearman | 3     | 0.001          | 0.013 | -0.030  | 0.032    |

Across generalization settings, the fingerprint MLP was slightly better on average in the pair, leave-drug, leave-cell-line, and scaffold evaluations. The corrected HDAC holdout was mixed by metric and every paired interval included zero. These tables replace all proxy state-of-the-art comparisons and do not support dual-stream superiority.

3 Candidate identity, condition coverage, and corrected holdout

Table S7: Candidate identity, BRD identifier, and frozen evidence role. Tianeptinaline/BG-1010 is retained only to document the identity conflict.

| Candidate                | BRD ID        | Frozen role                             | HDAC annotated | Name conflict | Interpretation                                   |
|--------------------------|---------------|-----------------------------------------|----------------|---------------|--------------------------------------------------|
| Mocetinostat             | BRD-K16485616 | core candidate                          | yes            | no            | Primary computational hypothesis                 |
| TC-H-106                 | BRD-K45044657 | exploratory / original-method-sensitive | yes            | no            | Limited HiQ coverage; close training neighbor    |
| NCH-51                   | BRD-K52522949 | secondary candidate                     | yes            | no            | Measured support; scaffold-exposure caveat       |
| Belinostat               | BRD-K17743125 | class support                           | yes            | no            | Measured HDAC-class support                      |
| PCI-24781                | BRD-K12867552 | class support                           | yes            | no            | Measured HDAC-class support                      |
| Panobinostat             | BRD-K02130563 | class support                           | yes            | no            | Measured HDAC-class support                      |
| Entinostat               | BRD-K77908580 | reference control                       | yes            | no            | Established HDAC reference                       |
| Vorinostat               | BRD-K81418486 | reference control                       | yes            | no            | Established HDAC reference                       |
| RG2833                   | BRD-K02389548 | prediction only                         | yes            | no            | No measured LINCS profile                        |
| Tianeptinaline / BG-1010 | BRD-A73972952 | identity conflict; excluded             | no             | yes           | Name-structure conflict; exact training exposure |

Table S8: Official LINCS condition and quality coverage. “All cells” and “HiQ cells” are deliberately separated.

| Candidate              | All | QC  | HiQ | All cells | HiQ cells | Times h     | Dose levels | Min $\mu$ M | Max $\mu$ M |
|------------------------|-----|-----|-----|-----------|-----------|-------------|-------------|-------------|-------------|
| Mocetinostat           | 14  | 8   | 8   | 12        | 8         | 6   24      | 1           | 10.000      | 10.000      |
| NCH-51                 | 17  | 8   | 8   | 10        | 8         | 6   24      | 1           | 10.000      | 10.000      |
| TC-H-106               | 6   | 4   | 4   | 5         | 4         | 6   24      | 1           | 10.000      | 10.000      |
| Belinostat             | 24  | 13  | 12  | 12        | 6         | 6   24      | 4           | 0.370       | 10.000      |
| PCI-24781              | 19  | 13  | 13  | 13        | 10        | 6   24      | 1           | 10.000      | 10.000      |
| Panobinostat           | 126 | 93  | 84  | 15        | 10        | 6   24      | 20          | 0.001       | 10.000      |
| Entinostat             | 101 | 77  | 67  | 14        | 10        | 6   24      | 13          | 0.040       | 10.000      |
| Vorinostat             | 899 | 521 | 487 | 54        | 35        | 6   24   48 | 38          | 0.001       | 30.000      |
| Tianeptinaline/BG-1010 | 6   | 1   | 1   | 2         | 1         | 6   24      | 1           | 10.000      | 10.000      |

The official LINCS cell count and the HiQ-eligible official-cell count are separate fields in Table S8. This resolves the earlier ambiguity in which a single “Cells” column could refer to either the full official metadata or the primary HiQ analysis.

Table S9: Pre-correction versus corrected annotation-defined HDAC holdout. The pre-correction row is provenance only and is excluded from inference.

| Version           | Test profiles | Test structures | Test cells | $\Delta$ profiles | $\Delta$ structures |
|-------------------|---------------|-----------------|------------|-------------------|---------------------|
| discarded_pre_fix | 1,565         | 21              | 55         | 0                 | 0                   |
| corrected_final   | 1,856         | 30              | 55         | 291               | 9                   |

The correction added nine annotation-defined structures and 291 profiles. NCH-51 was removed from training and placed into the final test set. Table S10 provides the complete compound-level inventory rather than only an aggregate count.

Table S10: Complete 30-structure corrected annotation-defined HDAC holdout inventory.

| BRD ID        | Compound        | Profiles | Cells | Annotation source  | Added by fix | NCH-51 removed |
|---------------|-----------------|----------|-------|--------------------|--------------|----------------|
| BRD-K85133207 | HDAC1-selective | 50       | 11    | metadata           | yes          | no             |
| BRD-K29313308 | HDAC3-selective | 55       | 11    | metadata           | yes          | no             |
| BRD-K76698671 | HNHA            | 8        | 7     | metadata           | yes          | no             |
| BRD-K69840642 | ISOX            | 48       | 43    | metadata           | yes          | no             |
| BRD-K83837640 | JNJ-26481585    | 11       | 9     | metadata_and_audit | no           | no             |
| BRD-K45528773 | M-344           | 6        | 3     | audit              | no           | no             |
| BRD-K04466929 | Merck60         | 51       | 8     | metadata_and_audit | no           | no             |
| BRD-K52522949 | NCH-51          | 17       | 11    | metadata           | yes          | yes            |
| BRD-K13169950 | NSC-3852        | 20       | 14    | metadata_and_audit | no           | no             |
| BRD-K12867552 | PCI-24781       | 19       | 13    | metadata_and_audit | no           | no             |
| BRD-K88742110 | PCI-34051       | 43       | 11    | metadata_and_audit | no           | no             |
| BRD-K45044657 | TC-H-106        | 6        | 5     | audit              | no           | no             |
| BRD-K74761218 | WT-171          | 15       | 11    | metadata           | yes          | no             |
| BRD-A63989068 | apicidin        | 6        | 3     | metadata           | yes          | no             |
| BRD-K64606589 | apicidin        | 85       | 14    | metadata           | yes          | no             |
| BRD-K17743125 | belinostat      | 24       | 12    | metadata_and_audit | no           | no             |
| BRD-K56957086 | dacinostat      | 27       | 12    | metadata_and_audit | no           | no             |
| BRD-K11558771 | droxinostat     | 9        | 8     | metadata_and_audit | no           | no             |
| BRD-K77908580 | entinostat      | 101      | 14    | metadata_and_audit | no           | no             |
| BRD-K13810148 | givinostat      | 5        | 4     | metadata_and_audit | no           | no             |
| BRD-K16485616 | mocetinostat    | 14       | 12    | metadata_and_audit | no           | no             |
| BRD-K02130563 | panobinostat    | 126      | 15    | metadata_and_audit | no           | no             |
| BRD-K67102207 | phenylbutyrate  | 6        | 6     | audit              | no           | no             |
| BRD-K11663430 | pyroxamide      | 13       | 10    | audit              | no           | no             |
| BRD-K22503835 | scriptaid       | 38       | 14    | metadata_and_audit | no           | no             |
| BRD-K52313696 | tacedinaline    | 83       | 11    | audit              | no           | no             |
| BRD-K98426715 | tubacin         | 7        | 6     | metadata           | yes          | no             |
| BRD-K00627859 | tubastatin-a    | 51       | 8     | metadata_and_audit | no           | no             |
| BRD-K41260949 | valproic-acid   | 13       | 6     | metadata_and_audit | no           | no             |
| BRD-K81418486 | vorinostat      | 899      | 55    | metadata_and_audit | no           | no             |

Table S11: Strict leave-drug candidate membership.

| Candidate              | Train | Test | Test cells | Strict LDO |
|------------------------|-------|------|------------|------------|
| Belinostat             | 24    | 0    | 0          | no         |
| Entinostat             | 101   | 0    | 0          | no         |
| Mocetinostat           | 0     | 14   | 12         | yes        |
| NCH-51                 | 17    | 0    | 0          | no         |
| PCI-24781              | 0     | 19   | 13         | yes        |
| Panobinostat           | 126   | 0    | 0          | no         |
| RG2833                 | 0     | 0    | 0          | no         |
| TC-H-106               | 6     | 0    | 0          | no         |
| Tianeptinaline/BG-1010 | 6     | 0    | 0          | no         |
| Vorinostat             | 899   | 0    | 0          | no         |

Supplementary Table S11 (continued). Strict leave-drug candidate metrics. Only Mocetinostat and PCI-24781 were true test candidates.

| Candidate    | Model           | Seeds | Profiles | Cells | Pearson | 95% low | 95% high | Spearman |
|--------------|-----------------|-------|----------|-------|---------|---------|----------|----------|
| Mocetinostat | dual_stream     | 3     | 14       | 12    | 0.483   | 0.393   | 0.574    | 0.476    |
| Mocetinostat | fingerprint_mlp | 3     | 14       | 12    | 0.504   | 0.468   | 0.539    | 0.499    |
| PCI-24781    | dual_stream     | 3     | 19       | 13    | 0.308   | 0.152   | 0.464    | 0.270    |
| PCI-24781    | fingerprint_mlp | 3     | 19       | 13    | 0.540   | 0.423   | 0.657    | 0.493    |

Only Mocetinostat and PCI-24781 satisfy strict leave-drug candidate membership. For PCI-24781, the seed-paired 95% interval favored the fingerprint baseline for profile correlations; no strict candidate result favored the dual stream. These three-seed intervals are uncertainty summaries rather than population-level significance tests.

## 4 Measured LINCS reversal and prediction–measurement concordance

Measured profiles were aggregated hierarchically: median within candidate–dose–time–official-cell condition, median across conditions within official cell, comparison with each cancer disease vector, and summary across official cells and cancers. The HiQ stratum is primary; QC-pass and all-profile rows are sensitivity analyses.

Table S12: Corrected-HDAC predicted versus HiQ measured reversal with 10,000 candidate–cancer crossed bootstrap iterations (seed 20260719).

| Metric            | Model           | Candidates | Cancers | Pearson | 95% low | 95% high | Spearman | 95% low | 95% high |
|-------------------|-----------------|------------|---------|---------|---------|----------|----------|---------|----------|
| signed_wtrs       | dual_stream     | 8          | 22      | 0.642   | 0.365   | 0.838    | 0.640    | 0.297   | 0.840    |
| signed_wtrs       | fingerprint_mlp | 8          | 22      | 0.805   | 0.601   | 0.911    | 0.803    | 0.554   | 0.919    |
| spearman_reversal | dual_stream     | 8          | 22      | 0.813   | 0.634   | 0.919    | 0.801    | 0.594   | 0.909    |
| spearman_reversal | fingerprint_mlp | 8          | 22      | 0.853   | 0.710   | 0.921    | 0.830    | 0.653   | 0.910    |

The crossed bootstrap independently resampled eight candidates and 22 cancers. This is more conservative than treating 176 candidate–cancer pairs as independent. Leave-one-candidate-out and leave-one-cancer-out influence tables are in Additional file 2 and retain positive associations.

Table S13: Primary official-condition HiQ measured reversal across 22 cancer signatures.

| Candidate    | Metric            | HiQ | HiQ cells | Median | Q1    | Q3    | Positive cancers % |
|--------------|-------------------|-----|-----------|--------|-------|-------|--------------------|
| Belinostat   | signed_wtrs       | 12  | 6         | 0.140  | 0.105 | 0.200 | 95.5               |
| Belinostat   | spearman_reversal | 12  | 6         | 0.058  | 0.037 | 0.092 | 95.5               |
| Entinostat   | signed_wtrs       | 67  | 10        | 0.082  | 0.041 | 0.101 | 95.5               |
| Entinostat   | spearman_reversal | 67  | 10        | 0.082  | 0.038 | 0.100 | 95.5               |
| Mocetinostat | signed_wtrs       | 8   | 8         | 0.262  | 0.172 | 0.316 | 95.5               |
| Mocetinostat | spearman_reversal | 8   | 8         | 0.137  | 0.088 | 0.151 | 95.5               |
| NCH-51       | signed_wtrs       | 8   | 8         | 0.194  | 0.144 | 0.249 | 95.5               |
| NCH-51       | spearman_reversal | 8   | 8         | 0.103  | 0.068 | 0.133 | 95.5               |
| PCI-24781    | signed_wtrs       | 13  | 10        | 0.139  | 0.111 | 0.211 | 95.5               |
| PCI-24781    | spearman_reversal | 13  | 10        | 0.075  | 0.057 | 0.107 | 95.5               |
| Panobinostat | signed_wtrs       | 84  | 10        | 0.191  | 0.152 | 0.224 | 95.5               |
| Panobinostat | spearman_reversal | 84  | 10        | 0.061  | 0.035 | 0.095 | 95.5               |
| TC-H-106     | signed_wtrs       | 4   | 4         | 0.119  | 0.081 | 0.160 | 90.9               |
| TC-H-106     | spearman_reversal | 4   | 4         | 0.113  | 0.071 | 0.158 | 90.9               |
| Vorinostat   | signed_wtrs       | 487 | 35        | 0.054  | 0.036 | 0.122 | 95.5               |
| Vorinostat   | spearman_reversal | 487 | 35        | 0.043  | 0.024 | 0.080 | 95.5               |

#### Seed-level corrected-HDAC prediction--measurement concordance

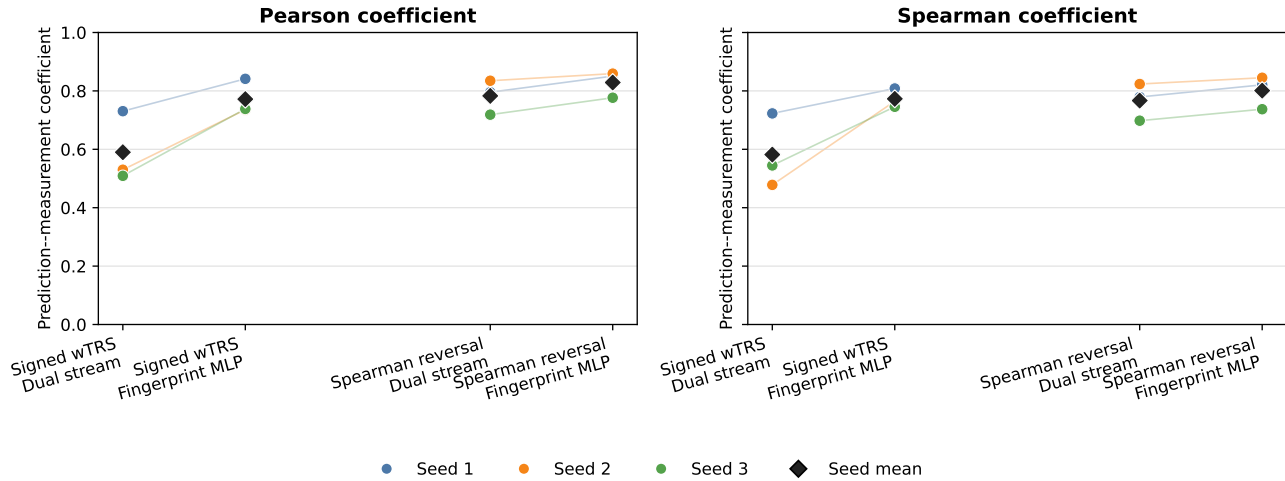
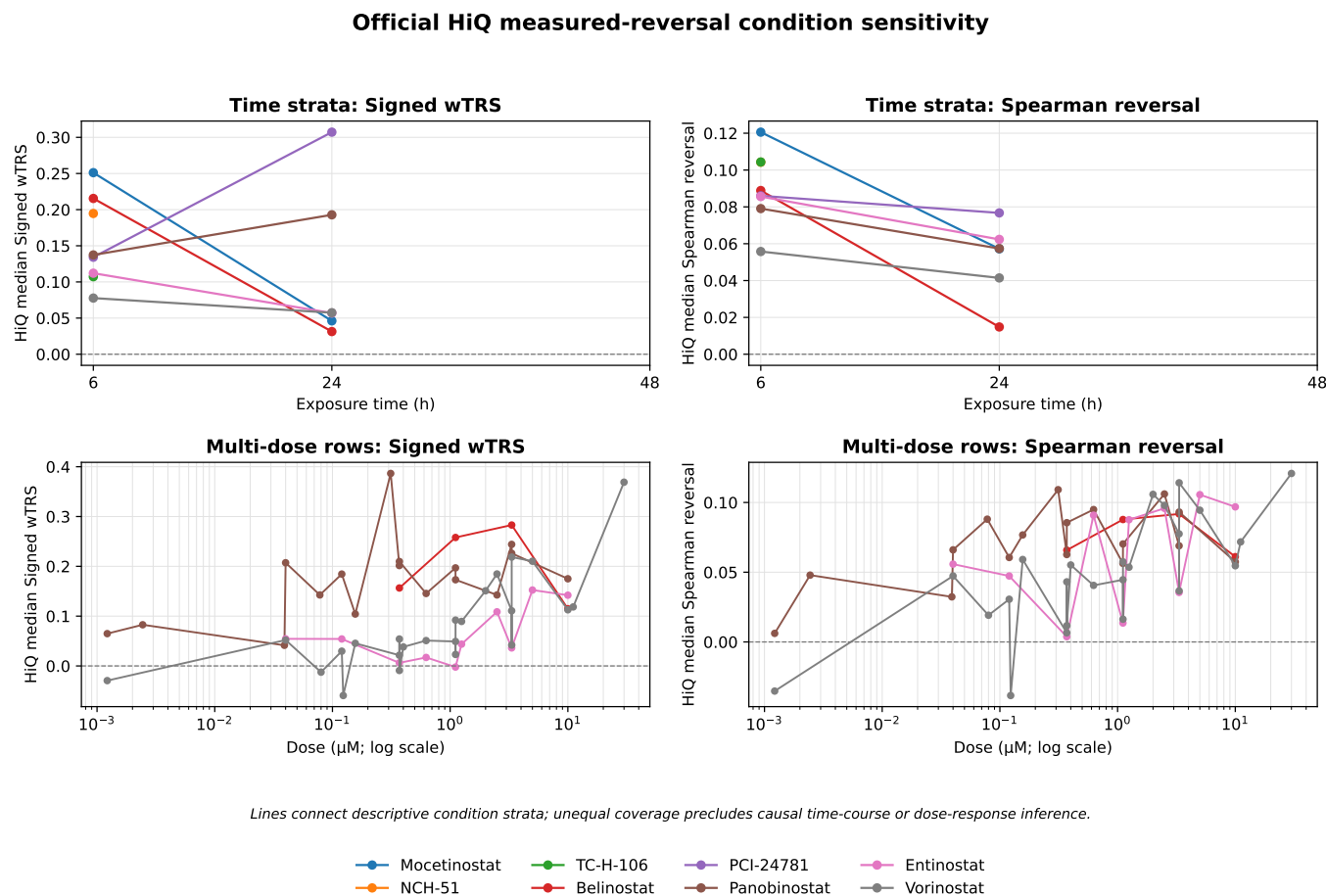


Figure S1: **Seed-level prediction–measurement concordance.** Pearson and Spearman coefficients are shown separately for every corrected annotation-defined HDAC seed, model, and primary reversal definition. Large markers denote seed means. The figure exposes run-to-run variability that is summarized by the crossed candidate–cancer intervals in the main text; it is not a second independent validation dataset.



**Figure S2: Measured LINCS time- and dose-stratum sensitivity.** HiQ pan-cancer median signed wTRS and Spearman reversal are shown by exposure time for the eight primary compounds and by dose for the four reference or class-support compounds with more than one profiled dose. Unequal cells, signatures, and condition coverage prevent causal dose-response or time-course interpretation; these panels are descriptive sensitivity checks.

All eight primary compounds had positive median measured reversal under signed and rank-based definitions. These are experimental expression responses, not direct drug-sensitivity, target-engagement, or efficacy measurements. The all-profile comparison against 11,445 canonical compounds in Additional file 2 is unmatched for dose, time, cell, and quality and is therefore not a condition-matched non-HDAC null.

## 5 Candidate stability, structural exposure, and formal class enrichment

The primary stability hierarchy contains 48 configurations: four generalization or distribution-shift regimes, two models, three seeds, and two primary reversal metrics. Adding the legacy reversal-only magnitude score produces 72 configurations and is explicitly sensitivity analysis. These regime labels do not imply that every named candidate is unseen under every configuration. Candidate identity and training exposure override descriptive rank; this is why the high-ranking identity-conflicted Tianeptinaline/BG-1010 row remains excluded.

Table S14: Candidate stability under the 48-configuration primary hierarchy and 72-configuration legacy-inclusive sensitivity analysis.

| Candidate              | Role                       | Primary n | Primary median | Primary worst | Sensitivity n | Sensitivity median | $\Delta$ primary-sens. |
|------------------------|----------------------------|-----------|----------------|---------------|---------------|--------------------|------------------------|
| Belinostat             | class_support              | 48        | 70.0           | 56.8          | 72            | 92.8               | -22.8                  |
| Entinostat             | reference_control          | 48        | 77.9           | 16.3          | 72            | 66.2               | 11.7                   |
| Mocetinostat           | core_candidate             | 48        | 92.2           | 53.8          | 72            | 91.9               | 0.3                    |
| NCH-51                 | secondary_candidate        | 48        | 88.5           | 65.6          | 72            | 95.4               | -6.9                   |
| PCI-24781              | class_support              | 48        | 57.6           | 32.3          | 72            | 57.6               | 0.0                    |
| Panobinostat           | class_support              | 48        | 60.6           | 33.5          | 72            | 94.2               | -33.5                  |
| RG2833                 | prediction_only            | 48        | 72.0           | 47.4          | 72            | 57.3               | 14.7                   |
| TC-H-106               | original_method_sensitive  | 48        | 84.3           | 49.1          | 72            | 74.7               | 9.6                    |
| Tianeptinaline/BG-1010 | identity_conflict_excluded | 48        | 90.0           | 71.0          | 72            | 76.7               | 13.3                   |
| Vorinostat             | reference_control          | 48        | 69.5           | 19.0          | 72            | 76.7               | -7.2                   |

Table S15: Corrected annotation-defined HDAC structural-neighbor and scaffold exposure audit.

| Candidate              | Role                                    | Train | Test | Max Tanimoto | Top-10 mean | Exact train | Scaffold train | Same-scaffold n | Conflict |
|------------------------|-----------------------------------------|-------|------|--------------|-------------|-------------|----------------|-----------------|----------|
| Mocetinostat           | core candidate                          | 0     | 14   | 0.537        | 0.447       | no          | no             | 0               | no       |
| TC-H-106               | exploratory / original-method-sensitive | 0     | 6    | 0.789        | 0.438       | no          | no             | 0               | no       |
| NCH-51                 | secondary candidate                     | 0     | 17   | 0.431        | 0.366       | no          | yes            | 1               | no       |
| Belinostat             | class support                           | 0     | 24   | 0.500        | 0.339       | no          | yes            | 8               | no       |
| PCI-24781              | class support                           | 0     | 19   | 0.410        | 0.317       | no          | no             | 0               | no       |
| Panobinostat           | class support                           | 0     | 126  | 0.381        | 0.315       | no          | yes            | 1               | no       |
| Entinostat             | reference control                       | 0     | 101  | 0.558        | 0.429       | no          | no             | 0               | no       |
| Vorinostat             | reference control                       | 0     | 899  | 0.697        | 0.503       | no          | yes            | 296             | no       |
| RG2833                 | prediction only                         | 0     | 0    | 0.574        | 0.407       | no          | no             | 0               | no       |
| Tianeptinaline/BG-1010 | identity conflict; excluded             | 6     | 0    | 1.000        | 0.508       | yes         | yes            | 1               | yes      |

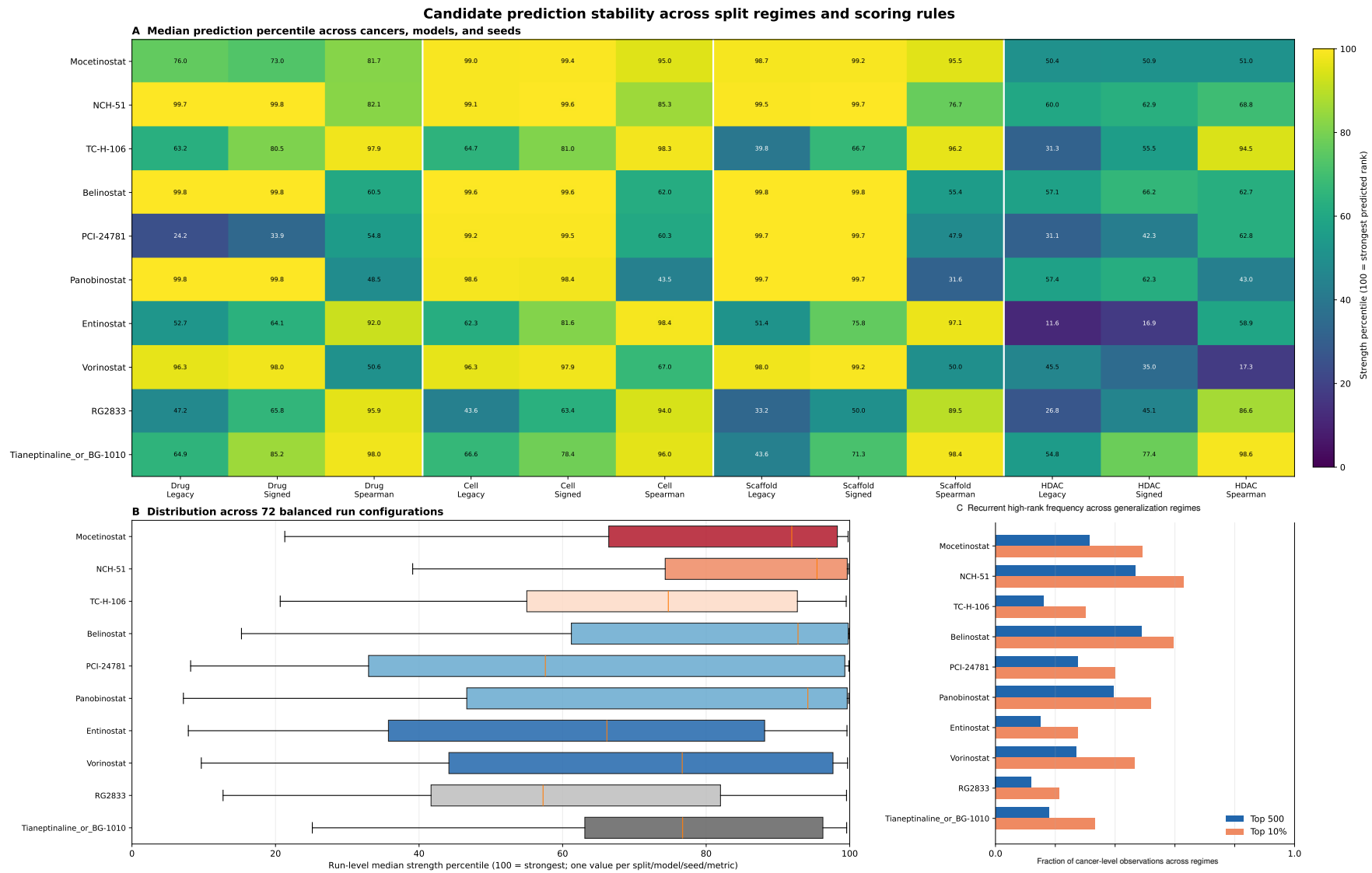


Figure S3: **Expanded split- and metric-level candidate stability sensitivity.** The heatmap reports the full 72-configuration legacy-inclusive audit by split and reversal definition, with run-level distributions and recurrent high-rank frequencies. This expands, rather than repeats, the 48-configuration primary summary in main Figure 4. Legacy wTRS remains a sensitivity metric and does not determine candidate eligibility.

Formal testing shows that the explicitly annotated class I subset is enriched under signed wTRS at every fixed revision cutoff, but not under Spearman reversal. For each metric, compounds were ordered by the mean of 528 equally weighted within-library rank fractions (24 screening runs across four generalization regimes, two models, and three seeds, multiplied by 22 cancers); 0 denotes the strongest rank, so lower mean rank fraction is better. Candidate stability uses a distinct strength-percentile orientation, where 100 denotes the strongest rank and higher is better. The cutoffs were fixed for the revision analysis but were not prospectively preregistered. The result is therefore metric-dependent rather than universal. The retained title refers to computational prioritization and does not imply model superiority, efficacy, or metric-independent enrichment. The broader HDAC definition and legacy metric are included in Additional file 2.

Table S16: Formal HDAC enrichment at fixed revision library cutoffs. Consensus ranks equally average 24 screening runs across 22 cancers per compound as mean rank fractions (0 = strongest); Fisher tests are one-sided and FDR is within each metric and annotation definition.

| Metric            | Definition              | Top fraction | Top k | Library HDAC | Observed | Expected | Fold  | <i>P</i> | FDR      |
|-------------------|-------------------------|--------------|-------|--------------|----------|----------|-------|----------|----------|
| signed_wtrs       | annotation_defined_HDAC | 0.5          | 143   | 39           | 4        | 0.20     | 20.42 | 4.37e-05 | 5.83e-05 |
| signed_wtrs       | annotation_defined_HDAC | 1.0          | 285   | 39           | 5        | 0.39     | 12.81 | 4.22e-05 | 5.83e-05 |
| signed_wtrs       | annotation_defined_HDAC | 5.0          | 1,424 | 39           | 11       | 1.95     | 5.64  | 2.16e-06 | 8.62e-06 |
| signed_wtrs       | annotation_defined_HDAC | 10.0         | 2,848 | 39           | 13       | 3.90     | 3.33  | 6.48e-05 | 6.48e-05 |
| signed_wtrs       | explicit_class_I_target | 0.5          | 143   | 26           | 4        | 0.13     | 30.64 | 8.36e-06 | 8.36e-06 |
| signed_wtrs       | explicit_class_I_target | 1.0          | 285   | 26           | 5        | 0.26     | 19.22 | 5.37e-06 | 7.16e-06 |
| signed_wtrs       | explicit_class_I_target | 5.0          | 1,424 | 26           | 11       | 1.30     | 8.46  | 1.81e-08 | 7.25e-08 |
| signed_wtrs       | explicit_class_I_target | 10.0         | 2,848 | 26           | 12       | 2.60     | 4.61  | 2.47e-06 | 4.93e-06 |
| spearman_reversal | annotation_defined_HDAC | 0.5          | 143   | 39           | 1        | 0.20     | 5.11  | 1.78e-01 | 3.50e-01 |
| spearman_reversal | annotation_defined_HDAC | 1.0          | 285   | 39           | 2        | 0.39     | 5.12  | 5.80e-02 | 2.32e-01 |
| spearman_reversal | annotation_defined_HDAC | 5.0          | 1,424 | 39           | 3        | 1.95     | 1.54  | 3.09e-01 | 3.50e-01 |
| spearman_reversal | annotation_defined_HDAC | 10.0         | 2,848 | 39           | 5        | 3.90     | 1.28  | 3.50e-01 | 3.50e-01 |
| spearman_reversal | explicit_class_I_target | 0.5          | 143   | 26           | 1        | 0.13     | 7.66  | 1.23e-01 | 4.60e-01 |
| spearman_reversal | explicit_class_I_target | 1.0          | 285   | 26           | 1        | 0.26     | 3.84  | 2.30e-01 | 4.60e-01 |
| spearman_reversal | explicit_class_I_target | 5.0          | 1,424 | 26           | 2        | 1.30     | 1.54  | 3.76e-01 | 4.90e-01 |
| spearman_reversal | explicit_class_I_target | 10.0         | 2,848 | 26           | 3        | 2.60     | 1.15  | 4.90e-01 | 4.90e-01 |

## 6 Frozen reversal-associated genes, pathways, and networks

Candidate roles were frozen before these analyses. At the 70% primary consensus threshold, the top 200 reversal-associated genes per candidate/control were carried forward. The term “reversal-associated” is used because gene selection derives from the same disease–perturbation opposition used for ranking; these genes are not direct pharmacological targets.

Table S17: Frozen reversal-associated gene definition and mapping coverage. Selection precedes network and pathway interpretation.

| Candidate    | Role                  | Threshold | Eligible | Stable direction | Top selected | STRING mappable | Mapping % |
|--------------|-----------------------|-----------|----------|------------------|--------------|-----------------|-----------|
| Mocetinostat | core_candidate        | 70.0      | 1,212    | 1,212            | 200          | 200             | 100.0     |
| NCH-51       | secondary_candidate   | 70.0      | 1,242    | 1,242            | 200          | 200             | 100.0     |
| TC-H-106     | exploratory_candidate | 70.0      | 1,324    | 1,324            | 200          | 200             | 100.0     |
| Belinostat   | class_support         | 70.0      | 1,167    | 1,167            | 200          | 200             | 100.0     |
| Entinostat   | reference_control     | 70.0      | 880      | 880              | 200          | 200             | 100.0     |
| Vorinostat   | reference_control     | 70.0      | 1,035    | 1,035            | 200          | 200             | 100.0     |

Table S18: Leading 20 reversal-associated genes per candidate or control by frozen consensus rank. These are not direct drug targets.

| Rank | Mocetinostat | NCH-51 | TC-H-106 | Belinostat | Entinostat | Vorinostat |
|------|--------------|--------|----------|------------|------------|------------|
| 1    | BIRC5        | UBE2C  | ADH1B    | UBE2C      | PDK4       | ATP1B2     |
| 2    | TCF21        | TCF21  | CES1     | RGN        | AKAP12     | FBLN5      |
| 3    | LYVE1        | GPX3   | BIRC5    | ATP1B2     | TEK        | CDT1       |
| 4    | FOSB         | FOSB   | CHRD1    | CDT1       | XPNPEP2    | MCM2       |
| 5    | TEK          | LYVE1  | UBE2C    | NR4A3      | FBLN5      | LMNB2      |
| 6    | NR4A3        | ATP1B2 | MYLK     | PTGDS      | CDK4       | GPX3       |
| 7    | GIN52        | MCM2   | PDK4     | MCM2       | FOX1       | CLN6       |
| 8    | RAD51        | AOX1   | AOX1     | FBLN5      | CDCA8      | TCF21      |
| 9    | PLK1         | LIFR   | GIN52    | POLE2      | FAM107A    | FOX1       |
| 10   | CCNA2        | CCDC69 | CDT1     | CDK4       | LEPR       | CCNB2      |
| 11   | GPM6A        | RGN    | ABCA8    | LMNB2      | RAD51AP1   | KIF4A      |
| 12   | HBB          | PDGFRA | CLEC3B   | TRIB3      | CCDC86     | AKAP12     |
| 13   | BUB1         | TXNIP  | PDGFRA   | DNAJB4     | FEN1       | TTK        |
| 14   | MCM10        | CENPM  | FOSB     | ZNF331     | DNAJC9     | NCAPH      |
| 15   | HJURP        | CDT1   | AQP7     | SLC25A20   | ADH1B      | ZNF282     |
| 16   | TPX2         | CCNB2  | PRELP    | CCNB2      | BIRC5      | SPC25      |
| 17   | CDCA3        | KIF20A | AKAP12   | PLK1       | UBE2C      | TRIB3      |
| 18   | ASF1B        | PLK1   | AOC3     | FOX1       | CHRD1      | ECT2       |
| 19   | KCNK3        | CCNA2  | FMO2     | CCNA2      | CES1       | PLK1       |
| 20   | CKAP2        | FOX1   | PRKAR2B  | GTSE1      | GPX3       | CDK4       |

Table S19: Scope of significant g:Profiler results by candidate, reversal direction, and ontology/source.

| Candidate    | Gene-set scope       | Source | Significant terms |
|--------------|----------------------|--------|-------------------|
| Belinostat   | all_selected         | GO:BP  | 167               |
| Belinostat   | all_selected         | GO:CC  | 44                |
| Belinostat   | all_selected         | GO:MF  | 4                 |
| Belinostat   | all_selected         | KEGG   | 2                 |
| Belinostat   | all_selected         | REAC   | 50                |
| Belinostat   | disease_down_drug_up | GO:BP  | 7                 |
| Belinostat   | disease_up_drug_down | GO:BP  | 272               |
| Belinostat   | disease_up_drug_down | GO:CC  | 75                |
| Belinostat   | disease_up_drug_down | GO:MF  | 68                |
| Belinostat   | disease_up_drug_down | KEGG   | 7                 |
| Belinostat   | disease_up_drug_down | REAC   | 100               |
| Entinostat   | all_selected         | GO:BP  | 182               |
| Entinostat   | all_selected         | GO:CC  | 56                |
| Entinostat   | all_selected         | GO:MF  | 31                |
| Entinostat   | all_selected         | KEGG   | 5                 |
| Entinostat   | all_selected         | REAC   | 93                |
| Entinostat   | disease_down_drug_up | GO:BP  | 7                 |
| Entinostat   | disease_down_drug_up | GO:CC  | 17                |
| Entinostat   | disease_down_drug_up | KEGG   | 1                 |
| Entinostat   | disease_down_drug_up | REAC   | 1                 |
| Entinostat   | disease_up_drug_down | GO:BP  | 281               |
| Entinostat   | disease_up_drug_down | GO:CC  | 111               |
| Entinostat   | disease_up_drug_down | GO:MF  | 106               |
| Entinostat   | disease_up_drug_down | KEGG   | 9                 |
| Entinostat   | disease_up_drug_down | REAC   | 146               |
| Mocetinostat | all_selected         | GO:BP  | 163               |
| Mocetinostat | all_selected         | GO:CC  | 30                |
| Mocetinostat | all_selected         | GO:MF  | 2                 |
| Mocetinostat | all_selected         | KEGG   | 2                 |
| Mocetinostat | all_selected         | REAC   | 51                |
| Mocetinostat | disease_down_drug_up | GO:BP  | 88                |
| Mocetinostat | disease_down_drug_up | GO:CC  | 13                |
| Mocetinostat | disease_down_drug_up | REAC   | 9                 |
| Mocetinostat | disease_up_drug_down | GO:BP  | 272               |
| Mocetinostat | disease_up_drug_down | GO:CC  | 88                |
| Mocetinostat | disease_up_drug_down | GO:MF  | 30                |
| Mocetinostat | disease_up_drug_down | KEGG   | 7                 |
| Mocetinostat | disease_up_drug_down | REAC   | 178               |
| NCH-51       | all_selected         | GO:BP  | 173               |
| NCH-51       | all_selected         | GO:CC  | 44                |
| NCH-51       | all_selected         | GO:MF  | 4                 |
| NCH-51       | all_selected         | KEGG   | 4                 |
| NCH-51       | all_selected         | REAC   | 47                |
| NCH-51       | disease_down_drug_up | GO:CC  | 3                 |
| NCH-51       | disease_down_drug_up | KEGG   | 1                 |
| NCH-51       | disease_down_drug_up | REAC   | 1                 |
| NCH-51       | disease_up_drug_down | GO:BP  | 302               |
| NCH-51       | disease_up_drug_down | GO:CC  | 91                |
| NCH-51       | disease_up_drug_down | GO:MF  | 69                |
| NCH-51       | disease_up_drug_down | KEGG   | 11                |
| NCH-51       | disease_up_drug_down | REAC   | 102               |
| TC-H-106     | all_selected         | GO:BP  | 12                |
| TC-H-106     | all_selected         | GO:CC  | 36                |
| TC-H-106     | all_selected         | REAC   | 4                 |
| TC-H-106     | disease_down_drug_up | GO:BP  | 21                |
| TC-H-106     | disease_down_drug_up | GO:CC  | 13                |
| TC-H-106     | disease_down_drug_up | KEGG   | 1                 |
| TC-H-106     | disease_up_drug_down | GO:BP  | 181               |
| TC-H-106     | disease_up_drug_down | GO:CC  | 83                |
| TC-H-106     | disease_up_drug_down | GO:MF  | 3                 |
| TC-H-106     | disease_up_drug_down | KEGG   | 3                 |
| TC-H-106     | disease_up_drug_down | REAC   | 66                |
| Vorinostat   | all_selected         | GO:BP  | 211               |
| Vorinostat   | all_selected         | GO:CC  | 69                |
| Vorinostat   | all_selected         | GO:MF  | 42                |
| Vorinostat   | all_selected         | KEGG   | 5                 |
| Vorinostat   | all_selected         | REAC   | 61                |
| Vorinostat   | disease_down_drug_up | GO:BP  | 12                |
| Vorinostat   | disease_down_drug_up | GO:MF  | 9                 |
| Vorinostat   | disease_up_drug_down | GO:BP  | 288               |
| Vorinostat   | disease_up_drug_down | GO:CC  | 93                |
| Vorinostat   | disease_up_drug_down | GO:MF  | 61                |
| Vorinostat   | disease_up_drug_down | KEGG   | 12                |
| Vorinostat   | disease_up_drug_down | REAC   | 124               |

Supplementary Table S19 (continued). Representative nonredundant enrichment terms after corrected gene-intersection redundancy filtering. Probabilities are shown in scientific notation rather than rounded to zero.

| Candidate    | Gene-set scope       | Source | Term ID            | Nonredundant term                                 | $P_{FDR}$ | Overlap | Term size | Effective domain |
|--------------|----------------------|--------|--------------------|---------------------------------------------------|-----------|---------|-----------|------------------|
| Belinostat   | disease_up_drug_down | GO:BP  | GO:0007059         | chromosome segregation                            | 1.61e-29  | 36      | 294       | 11,154           |
| Belinostat   | disease_up_drug_down | GO:BP  | GO:0010965         | regulation of mitotic sister chromatid separation | 2.36e-18  | 15      | 47        | 11,154           |
| Belinostat   | disease_up_drug_down | GO:CC  | GO:0098687         | chromosomal region                                | 1.51e-16  | 25      | 283       | 11,154           |
| Belinostat   | disease_up_drug_down | GO:CC  | GO:0005819         | spindle                                           | 2.68e-14  | 23      | 305       | 11,154           |
| Belinostat   | disease_up_drug_down | GO:MF  | GO:0008017         | microtubule binding                               | 1.45e-07  | 14      | 182       | 11,154           |
| Belinostat   | disease_up_drug_down | GO:MF  | GO:0120543         | macromolecular conformation isomerase activity    | 3.75e-05  | 12      | 217       | 11,154           |
| Belinostat   | disease_up_drug_down | KEGG   | KEGG:04110         | Cell cycle                                        | 3.28e-10  | 14      | 130       | 11,154           |
| Belinostat   | disease_up_drug_down | KEGG   | KEGG:03030         | DNA replication                                   | 4.93e-07  | 7       | 34        | 11,154           |
| Belinostat   | disease_up_drug_down | REAC   | REAC:R-HSA-1640170 | Cell Cycle                                        | 1.29e-30  | 43      | 505       | 11,154           |
| Belinostat   | disease_up_drug_down | REAC   | REAC:R-HSA-69620   | Cell Cycle Checkpoints                            | 4.53e-15  | 21      | 209       | 11,154           |
| Entinostat   | disease_up_drug_down | GO:BP  | GO:1903047         | mitotic cell cycle process                        | 1.21e-16  | 37      | 568       | 11,154           |
| Entinostat   | disease_up_drug_down | GO:BP  | GO:0022613         | ribonucleoprotein complex biogenesis              | 1.77e-13  | 26      | 333       | 11,154           |
| Entinostat   | disease_up_drug_down | GO:CC  | GO:0098687         | chromosomal region                                | 1.94e-18  | 29      | 283       | 11,154           |
| Entinostat   | disease_up_drug_down | GO:CC  | GO:0000940         | outer kinetochore                                 | 5.51e-10  | 7       | 13        | 11,154           |
| Entinostat   | disease_up_drug_down | GO:MF  | GO:0003723         | RNA binding                                       | 3.72e-12  | 44      | 1,188     | 11,154           |
| Entinostat   | disease_up_drug_down | GO:MF  | GO:0140640         | catalytic activity, acting on a nucleic acid      | 3.72e-12  | 27      | 410       | 11,154           |
| Entinostat   | disease_up_drug_down | KEGG   | KEGG:04110         | Cell cycle                                        | 6.12e-11  | 16      | 130       | 11,154           |
| Entinostat   | disease_up_drug_down | KEGG   | KEGG:03030         | DNA replication                                   | 1.41e-07  | 8       | 34        | 11,154           |
| Entinostat   | disease_up_drug_down | REAC   | REAC:R-HSA-1640170 | Cell Cycle                                        | 2.80e-24  | 42      | 505       | 11,154           |
| Entinostat   | disease_up_drug_down | REAC   | REAC:R-HSA-69306   | DNA Replication                                   | 7.78e-11  | 15      | 111       | 11,154           |
| Mocetinostat | disease_up_drug_down | GO:BP  | GO:1903047         | mitotic cell cycle process                        | 7.66e-30  | 41      | 568       | 11,154           |
| Mocetinostat | disease_up_drug_down | GO:BP  | GO:0000070         | mitotic sister chromatid segregation              | 7.46e-18  | 19      | 142       | 11,154           |
| Mocetinostat | disease_up_drug_down | GO:CC  | GO:0000793         | condensed chromosome                              | 1.12e-14  | 19      | 197       | 11,154           |
| Mocetinostat | disease_up_drug_down | GO:CC  | GO:0005819         | spindle                                           | 4.62e-13  | 20      | 305       | 11,154           |
| Mocetinostat | disease_up_drug_down | GO:MF  | GO:0008017         | microtubule binding                               | 1.34e-05  | 11      | 182       | 11,154           |
| Mocetinostat | disease_up_drug_down | GO:MF  | GO:0016853         | isomerase activity                                | 4.59e-03  | 10      | 324       | 11,154           |
| Mocetinostat | disease_up_drug_down | KEGG   | KEGG:04110         | Cell cycle                                        | 1.09e-06  | 10      | 130       | 11,154           |
| Mocetinostat | disease_up_drug_down | KEGG   | KEGG:04114         | Oocyte meiosis                                    | 9.33e-04  | 6       | 95        | 11,154           |
| Mocetinostat | disease_up_drug_down | REAC   | REAC:R-HSA-69278   | Cell Cycle, Mitotic                               | 4.71e-25  | 33      | 405       | 11,154           |
| Mocetinostat | disease_up_drug_down | REAC   | REAC:R-HSA-69618   | Mitotic Spindle Checkpoint                        | 3.78e-11  | 12      | 86        | 11,154           |
| NCH-51       | disease_up_drug_down | GO:BP  | GO:1903047         | mitotic cell cycle process                        | 5.48e-33  | 46      | 568       | 11,154           |
| NCH-51       | all_selected         | GO:BP  | GO:0007059         | chromosome segregation                            | 8.04e-19  | 38      | 294       | 11,154           |
| NCH-51       | disease_up_drug_down | GO:CC  | GO:0000793         | condensed chromosome                              | 3.11e-17  | 22      | 197       | 11,154           |
| NCH-51       | disease_up_drug_down | GO:CC  | GO:0005819         | spindle                                           | 4.20e-16  | 24      | 305       | 11,154           |
| NCH-51       | disease_up_drug_down | GO:MF  | GO:0008017         | microtubule binding                               | 5.17e-08  | 14      | 182       | 11,154           |
| NCH-51       | disease_up_drug_down | GO:MF  | GO:0120543         | macromolecular conformation isomerase activity    | 1.58e-05  | 12      | 217       | 11,154           |
| NCH-51       | disease_up_drug_down | KEGG   | KEGG:04110         | Cell cycle                                        | 2.06e-09  | 13      | 130       | 11,154           |
| NCH-51       | disease_up_drug_down | KEGG   | KEGG:03030         | DNA replication                                   | 2.85e-07  | 7       | 34        | 11,154           |
| NCH-51       | disease_up_drug_down | REAC   | REAC:R-HSA-69278   | Cell Cycle, Mitotic                               | 2.13e-30  | 39      | 405       | 11,154           |
| NCH-51       | disease_up_drug_down | REAC   | REAC:R-HSA-68886   | M Phase                                           | 4.29e-11  | 19      | 279       | 11,154           |
| TC-H-106     | disease_up_drug_down | GO:BP  | GO:0022613         | ribonucleoprotein complex biogenesis              | 3.48e-12  | 22      | 333       | 11,154           |
| TC-H-106     | disease_up_drug_down | GO:BP  | GO:0006260         | DNA replication                                   | 1.39e-06  | 13      | 213       | 11,154           |
| TC-H-106     | disease_up_drug_down | GO:CC  | GO:1990904         | ribonucleoprotein complex                         | 2.01e-07  | 19      | 510       | 11,154           |

Continued on next page

Table continued

| Candidate  | Gene-set scope       | Source | Term ID            | Nonredundant term                                 | $P_{FDR}$ | Overlap | Term size | Effective domain |
|------------|----------------------|--------|--------------------|---------------------------------------------------|-----------|---------|-----------|------------------|
| TC-H-106   | disease_up_drug_down | GO:CC  | GO:0005730         | nucleolus                                         | 6.61e-07  | 22      | 751       | 11,154           |
| TC-H-106   | disease_up_drug_down | GO:MF  | GO:0003723         | RNA binding                                       | 1.17e-08  | 32      | 1,188     | 11,154           |
| TC-H-106   | disease_up_drug_down | GO:MF  | GO:0016887         | ATP hydrolysis activity                           | 1.12e-02  | 10      | 305       | 11,154           |
| TC-H-106   | disease_up_drug_down | KEGG   | KEGG:03040         | Spliceosome                                       | 1.60e-04  | 8       | 105       | 11,154           |
| TC-H-106   | disease_up_drug_down | KEGG   | KEGG:03030         | DNA replication                                   | 8.24e-03  | 4       | 34        | 11,154           |
| TC-H-106   | disease_up_drug_down | REAC   | REAC:R-HSA-69278   | Cell Cycle, Mitotic                               | 2.53e-06  | 17      | 405       | 11,154           |
| TC-H-106   | disease_up_drug_down | REAC   | REAC:R-HSA-8953854 | Metabolism of RNA                                 | 1.19e-05  | 18      | 553       | 11,154           |
| Vorinostat | disease_up_drug_down | GO:BP  | GO:0007059         | chromosome segregation                            | 2.31e-25  | 36      | 294       | 11,154           |
| Vorinostat | disease_up_drug_down | GO:BP  | GO:0010965         | regulation of mitotic sister chromatid separation | 8.87e-17  | 15      | 47        | 11,154           |
| Vorinostat | disease_up_drug_down | GO:CC  | GO:0098687         | chromosomal region                                | 9.27e-22  | 32      | 283       | 11,154           |
| Vorinostat | all_selected         | GO:CC  | GO:0005694         | chromosome                                        | 7.93e-16  | 72      | 1,361     | 11,154           |
| Vorinostat | disease_up_drug_down | GO:MF  | GO:0120543         | macromolecular conformation isomerase activity    | 1.88e-06  | 15      | 217       | 11,154           |
| Vorinostat | disease_up_drug_down | GO:MF  | GO:0140640         | catalytic activity, acting on a nucleic acid      | 1.88e-06  | 20      | 410       | 11,154           |
| Vorinostat | disease_up_drug_down | KEGG   | KEGG:04110         | Cell cycle                                        | 5.12e-11  | 16      | 130       | 11,154           |
| Vorinostat | disease_up_drug_down | KEGG   | KEGG:03030         | DNA replication                                   | 2.99e-06  | 7       | 34        | 11,154           |
| Vorinostat | disease_up_drug_down | REAC   | REAC:R-HSA-1640170 | Cell Cycle                                        | 4.00e-25  | 43      | 505       | 11,154           |
| Vorinostat | disease_up_drug_down | REAC   | REAC:R-HSA-2500257 | Resolution of Sister Chromatid Cohesion           | 1.25e-11  | 15      | 98        | 11,154           |

The g:Profiler request submitted 11,144 unique measured Entrez identifiers as the custom background. The service response reported an effective mapped statistical domain of 11,154. These values answer different questions and are both retained. For every term, the returned per-query evidence array was aligned to the mapped-query order in the frozen response; nonempty entries identify the intersecting Entrez genes, while GO evidence codes are retained in a separate machine-readable field. All 5,037 reconstructed gene intersections match the service-reported intersection sizes. Table S19 reports candidate/source scope and de-duplicates representative term names;  $P$  values remain in scientific notation rather than appearing as “0.00.” Additional file 2 contains the complete term table and a cache/hash audit.

Table S20: STRING physical and functional network scope at required score 700.

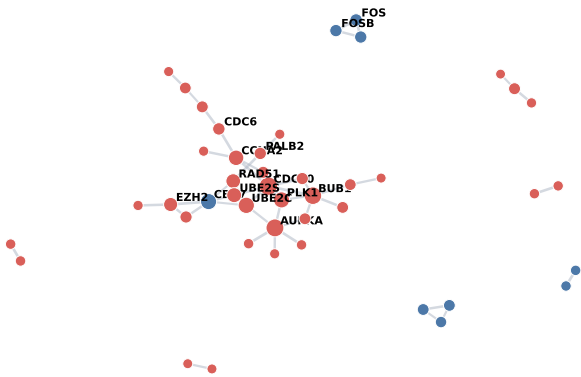
| Candidate    | Network    | Selected | Nodes with edges | Edges | Density | Components | Largest component |
|--------------|------------|----------|------------------|-------|---------|------------|-------------------|
| Mocetinostat | physical   | 200      | 49               | 52    | 0.003   | 160        | 30                |
| Mocetinostat | functional | 200      | 110              | 577   | 0.029   | 102        | 66                |
| NCH-51       | physical   | 200      | 56               | 59    | 0.003   | 156        | 25                |
| NCH-51       | functional | 200      | 116              | 944   | 0.047   | 98         | 86                |
| TC-H-106     | physical   | 200      | 43               | 52    | 0.003   | 167        | 13                |
| TC-H-106     | functional | 200      | 88               | 243   | 0.012   | 121        | 65                |
| Belinostat   | physical   | 200      | 55               | 61    | 0.003   | 155        | 27                |
| Belinostat   | functional | 200      | 105              | 968   | 0.049   | 104        | 81                |
| Entinostat   | physical   | 200      | 65               | 117   | 0.006   | 142        | 38                |
| Entinostat   | functional | 200      | 118              | 673   | 0.034   | 92         | 59                |
| Vorinostat   | physical   | 200      | 63               | 106   | 0.005   | 147        | 31                |
| Vorinostat   | functional | 200      | 122              | 918   | 0.046   | 87         | 95                |

Supplementary Table S20 (continued). Primary-candidate overlap with Entinostat and Vorinostat reversal-associated gene sets.

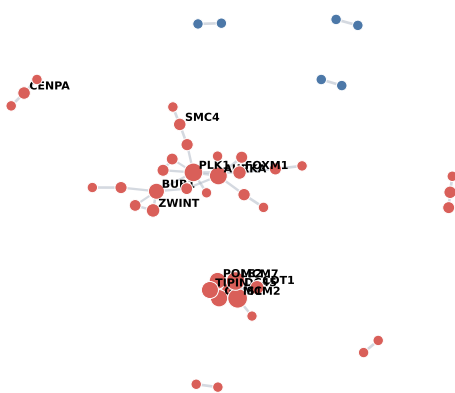
| Candidate    | Reference  | Genes A | Genes B | Shared | Jaccard |
|--------------|------------|---------|---------|--------|---------|
| Mocetinostat | Entinostat | 200     | 200     | 86     | 0.274   |
| Mocetinostat | Vorinostat | 200     | 200     | 63     | 0.187   |
| NCH-51       | Entinostat | 200     | 200     | 74     | 0.227   |
| NCH-51       | Vorinostat | 200     | 200     | 100    | 0.333   |
| TC-H-106     | Entinostat | 200     | 200     | 98     | 0.325   |
| TC-H-106     | Vorinostat | 200     | 200     | 62     | 0.183   |

Expanded high-confidence STRING physical networks of frozen candidates and controls

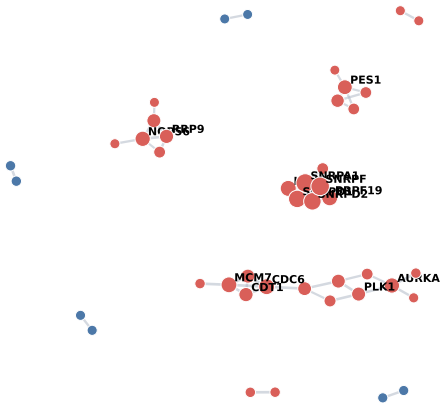
Mocetinostat: 45 non-isolated nodes, 49 displayed edges



NCH-51: 45 non-isolated nodes, 51 displayed edges



TC-H-106: 43 non-isolated nodes, 52 displayed edges



Interpretation key

Red: disease-up / drug-down reversal-associated gene

Blue: disease-down / drug-up reversal-associated gene

Node size: physical-network weighted degree

Edge width: STRING physical score

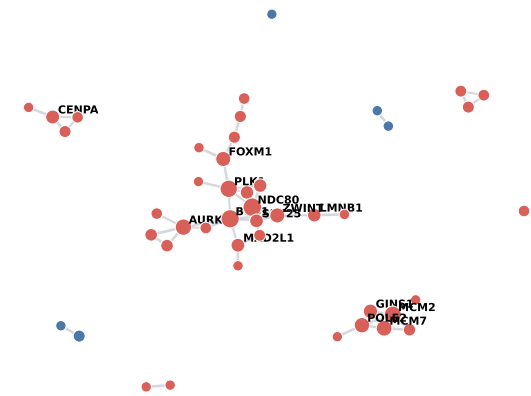
Labels: leading hubs only

Displayed genes are expanded plotting subsets.  
Full 200-gene sets and all physical/functional edges  
are supplied in Additional file 2.

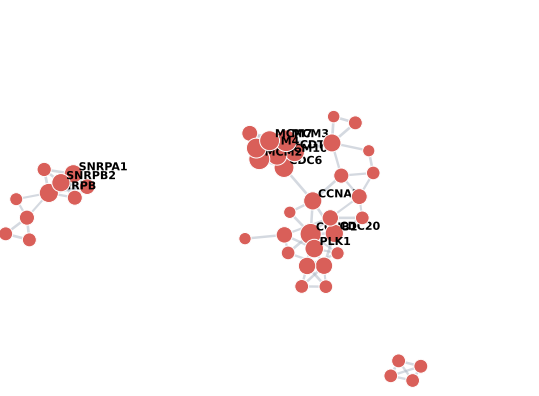
Figure S4: (A) Expanded high-confidence physical STRING plotting subsets. Mocetinostat, NCH-51, and TC-H-106 are shown with up to 45 non-isolated nodes selected by weighted degree. Leading hubs are labeled. Full 200-gene node tables and all physical/functional edges are in Additional file 2.

Expanded high-confidence STRING physical networks of frozen candidates and controls

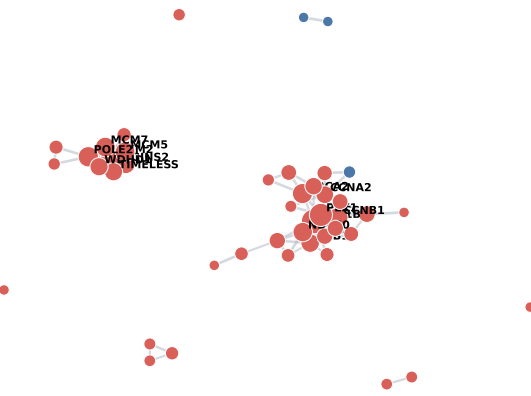
Belinostat: 45 non-isolated nodes, 52 displayed edges



Entinostat: 45 non-isolated nodes, 95 displayed edges



Vorinostat: 45 non-isolated nodes, 90 displayed edges



Interpretation key

Red: disease-up / drug-down reversal-associated gene

Blue: disease-down / drug-up reversal-associated gene

Node size: physical-network weighted degree  
Edge width: STRING physical score  
Labels: leading hubs only

Displayed genes are expanded plotting subsets.  
Full 200-gene sets and all physical/functional edges  
are supplied in Additional file 2.

Supplementary Figure S4 (continued), panel B. Expanded high-confidence physical STRING plotting subsets for controls. Belinostat, Entinostat, and Vorinostat are shown under the same plotting policy. Nodes remain reversal-associated genes, not direct drug targets.

7 DepMap genetic-dependency context

Table S21: DepMap Public 26Q1 pan-cancer class I HDAC gene-effect summary.

| Gene  | Models | Median | 95% low | 95% high | $\leq -0.5$ % | $\leq -1.0$ % |
|-------|--------|--------|---------|----------|---------------|---------------|
| HDAC1 | 1,208  | -0.107 | -0.118  | -0.093   | 4.5           | 1.1           |
| HDAC2 | 1,208  | 0.001  | -0.008  | 0.013    | 3.1           | 0.5           |
| HDAC3 | 1,208  | -1.215 | -1.249  | -1.188   | 95.1          | 68.2          |
| HDAC8 | 1,208  | -0.121 | -0.134  | -0.107   | 4.9           | 0.1           |

Supplementary Table S21 (continued). Lineage heterogeneity. Effect sizes are modest despite FDR significance.

| Gene  | Lineages | Models | Kruskal $H$ | $P$      | $\epsilon^2$ | FDR      |
|-------|----------|--------|-------------|----------|--------------|----------|
| HDAC1 | 24       | 1,193  | 107.32      | 7.49e-13 | 0.072        | 1.50e-12 |
| HDAC2 | 24       | 1,193  | 119.95      | 4.24e-15 | 0.083        | 1.70e-14 |
| HDAC3 | 24       | 1,193  | 95.64       | 7.90e-11 | 0.062        | 1.05e-10 |
| HDAC8 | 24       | 1,193  | 71.71       | 6.61e-07 | 0.042        | 6.61e-07 |

HDAC3 is the dominant pan-cancer dependency in DepMap Public 26Q1. The lineage tests are statistically significant with modest effect sizes, and pairwise HDAC correlations are weak. These results provide genetic-loss context but do not identify a candidate’s pharmacological isoform, efficacy, or normal-tissue safety.

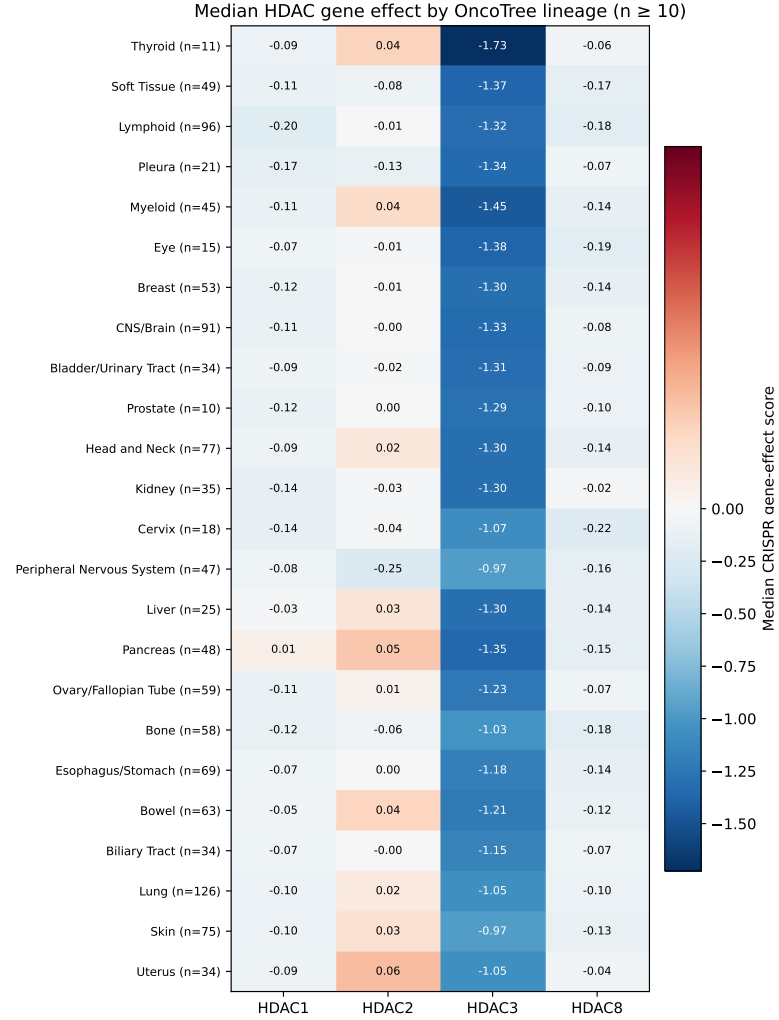


Figure S5: **DepMap lineage heterogeneity.** Median class I HDAC CRISPR gene effects are shown only for lineages meeting the prespecified minimum of ten models. More negative values indicate stronger loss-of-function effect.

## 8 Controlled docking and receptor sensitivity

Table S22: 4LXZ crystallographic redocking control. The prespecified RMSD gate was 2 Angstrom.

| Method        | Seeds | Seed coverage | Cluster poses | Median score | Medoid RMSD | Zn min | Zn max |
|---------------|-------|---------------|---------------|--------------|-------------|--------|--------|
| vina_standard | 3     | 3             | 8             | -7.290       | 4.771       | 2.348  | 3.645  |
| ad4zn         | 3     | 3             | 3             | -7.491       | 1.091       | 2.123  | 3.108  |

Supplementary Table S22 (continued). Frozen 4LXZ control/candidate panel. Favorable score is not evidence of intended zinc geometry.

| Compound                  | Role                                        | 3-seed | Poses | Cluster score | Medoid score | ZBG min | ZBG max | Nearest heteroatom |
|---------------------------|---------------------------------------------|--------|-------|---------------|--------------|---------|---------|--------------------|
| Vorinostat                | crystallographic_positive_control           | yes    | 3     | -7.491        | -7.532       | 2.123   | 3.108   | 2.123              |
| Vorinostat O-methyl decoy | designed zinc-chelation sensitivity control | yes    | 6     | -7.998        | -7.791       | 6.799   | 8.992   | 4.078              |
| Mocetinostat              | core_candidate                              | yes    | 7     | -11.024       | -12.289      | 12.954  | 13.872  | 2.002              |
| NCH-51                    | secondary_candidate                         | yes    | 8     | -8.357        | -9.094       | —       | —       | 2.448              |
| TC-H-106                  | original_method_sensitive_candidate         | yes    | 5     | -9.674        | -9.895       | 10.442  | 13.748  | 3.858              |
| Entinostat                | reference_control                           | yes    | 10    | -10.703       | -11.996      | 11.852  | 13.867  | 1.918              |
| Belinostat                | class_support_control                       | yes    | 3     | -9.336        | -9.383       | 8.918   | 9.525   | 2.841              |

Supplementary Table S22 (continued). HDAC2 4LXZ versus aligned HDAC1 4BKX receptor sensitivity.

| Compound                  | Pose RMSD | 4LXZ score | 4BKX score | $\Delta$ score | 4LXZ nearest Zn | 4BKX nearest Zn |
|---------------------------|-----------|------------|------------|----------------|-----------------|-----------------|
| Vorinostat                | 8.357     | -7.491     | -8.111     | -0.620         | 2.123           | 2.324           |
| Vorinostat O-methyl decoy | 1.872     | -7.998     | -7.767     | 0.231          | 4.078           | 2.210           |
| Mocetinostat              | 1.080     | -11.024    | -10.864    | 0.160          | 2.002           | 2.176           |
| NCH-51                    | 1.870     | -8.357     | -8.457     | -0.100         | 2.448           | 2.435           |
| TC-H-106                  | 1.508     | -9.674     | -8.886     | 0.788          | 3.858           | 3.895           |
| Entinostat                | 1.280     | -10.703    | -10.539    | 0.164          | 1.918           | 1.907           |
| Belinostat                | 8.787     | -9.336     | -7.915     | 1.421          | 2.841           | 2.140           |

The redocking control establishes protocol suitability only for one known cocrystal geometry under the zinc-aware potential. The O-methyl decoy demonstrates that a favorable score can coexist with an implausible prespecified zinc-binding-group distance. Several candidate poses change across 4LXZ and 4BKX. Consequently, docking is a protocol/receptor-sensitivity analysis with unresolved pose plausibility, not binding validation.

## Pose-level 4LXZ crystallographic redocking diagnostics

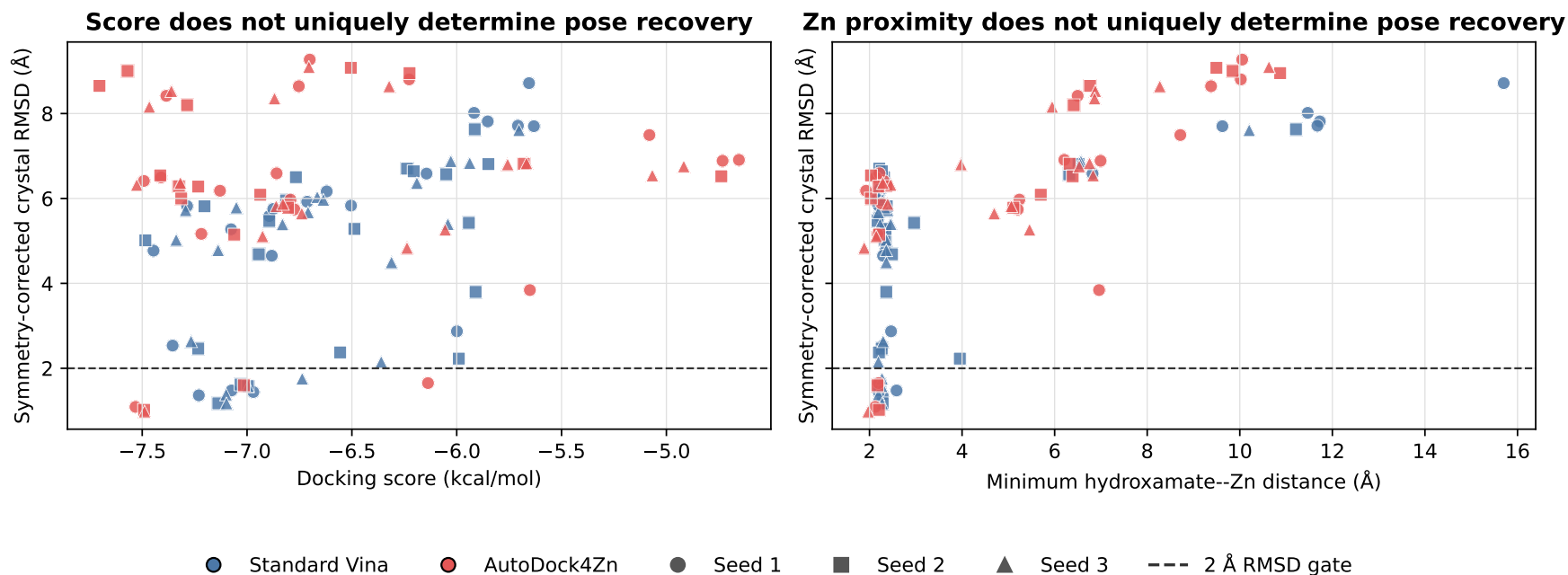


Figure S6: **Pose-level crystallographic redocking diagnostics.** Every retained pose from seeds 1–3 is shown by docking method. Score, symmetry-corrected heavy-atom RMSD, and hydroxamate–zinc distance are displayed jointly with the prespecified 2 Å RMSD gate. These pose-level diagnostics expand main Figure 9 and show why score or zinc proximity alone cannot establish a recovered binding geometry.

## 9 External drug-sensitivity availability and reproducibility

Table S23: External direct drug-sensitivity availability and null results. The public-transcriptomic option was completed; no surrogate is presented as direct validation.

| Scope                 | Endpoint                                | Availability                                                                         | Result                                                                        | Interpretation                                         |
|-----------------------|-----------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------|
| TC-H-106              | Direct public viability response        | Unavailable in frozen public-data audit                                              | Not performed                                                                 | No direct external drug-sensitivity validation claimed |
| Entinostat            | GDSC surrogate correlation              | Available but not candidate matched                                                  | Pearson R=-0.052; P=0.859                                                     | Null result; cannot validate TC-H-106                  |
| All frozen candidates | New wet-lab viability/biochemical assay | Not performed; reviewer suggested wet-lab assays or public transcriptomic validation | Official-condition measured<br>LINCS used as the public transcriptomic option | Does not replace viability or biochemical validation   |

The non-significant Entinostat surrogate result is retained as a null result and cannot validate TC-H-106. No direct candidate-matched public viability response was found for TC-H-106 in the frozen audit. This limitation is stated in the manuscript and response letter rather than replaced with a post hoc proxy.

Table S24: Reproducibility inventory. Unrecorded CPU/RAM values are disclosed rather than guessed.

| Item                         | Frozen value                                                                       | Provenance/limitation                                                         |
|------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Neural hardware              | NVIDIA GeForce RTX 4090                                                            | Recorded by run manifests                                                     |
| CPU model                    | Not captured in frozen AutoDL manifests                                            | Not reconstructed or fabricated                                               |
| RAM                          | Not captured in frozen AutoDL manifests                                            | Not reconstructed or fabricated                                               |
| Python                       | 3.12.3                                                                             | Run manifests                                                                 |
| PyTorch/CUDA                 | 2.5.1+cu124                                                                        | Run manifests                                                                 |
| RDKit                        | 2025.03.6 for docking; modeling canonicalization version in repository environment | Environment manifests                                                         |
| Neural benchmark run-time    | 5,112.6 s summed across 34 reported runs                                           | Run-level values in Table S5                                                  |
| Docking runtimes             | 330.0 s redocking; 2,121.9 s 4LXZ panel; 2,134.9 s 4BXX sensitivity                | CPU workflows                                                                 |
| Split seed                   | 42                                                                                 | Top-level grouping                                                            |
| Run seeds                    | 1–5 pair; 1–3 other splits                                                         | Frozen benchmark                                                              |
| Crossed-bootstrap seed       | 20260719                                                                           | 10,000 iterations                                                             |
| Repository evidence snapshot | fefbf2f4fa7399983d4d4041dcb8e7e91b84d17f                                           | Corrected statistics, scripts, exclusions, and tracker on major-revision-2026 |

CPU model and RAM were not captured in the frozen AutoDL manifests. They are explicitly marked unrecorded rather than inferred from the current environment. The corrected evidence snapshot is archived at repository commit `fefbf2f4fa7399983d4d4041dcb8e7e91b84d17f`; later document-only commits do not alter that frozen analysis snapshot.

## 10 Machine-readable companion

Additional file 2 contains the complete preserved run-level, candidate-level, network, pathway, DepMap, docking, sensitivity, and reproducibility tables with filters and a data dictionary. A TCGA cohort manifest identifies all 22 GDC projects and the frozen disease-matrix hashes; per-cohort final tumor/normal counts were not retained and are not retrospectively reconstructed. A separate condition-level manifest records the schema of the analysis-time 83,490-row table and explicitly discloses that the source file and its SHA256 were not preserved in the frozen portable package. All preserved aggregates used for the reported figures and inferences are included. Repository-relative source paths are used; no AutoDL absolute paths are required to interpret the workbook. Blank fields are retained only when a measurement is not applicable or was not recorded, and the corresponding reason is supplied in a status or note column.